QUBO Formulation: Clinical Enzyme Target Identification Problem

Clinical Enzyme Target Identification Problem

Problem Statement. Let $\mathcal{E}$ denote the set of enzyme candidates with cardinality $N = |\mathcal{E}|$. For each enzyme $i \in \mathcal{E}$, let $b_i \geq 0$ represent the target benefit score associated with selecting enzyme i. For every pair of distinct enzymes $i, j \in \mathcal{E}$ with $i < j$, let $p_{ij} \geq 0$ represent the redundancy penalty incurred if both enzymes i and j are selected. The penalties are symmetric, i.e., $p_{ij} = p_{ji}$, and $p_{ii} = 0$ for all i. The objective is to select a subset of exactly K enzymes such that the total benefit score of the selected enzymes minus the sum of redundancy penalties between all selected pairs is maximized.

Decision Variables. For each enzyme $i \in \mathcal{E}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if enzyme i is selected as a target, and 0 otherwise. The solution is represented by the vector $x = (x_i)_{i \in \mathcal{E}} \in \{0, 1\}^N$.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) formulation by converting the maximization objective into a minimization problem and incorporating the cardinality constraint via a penalty term.

Step 1 – Objective Function. The original objective is to maximize the total score:

$$\max_{x \in \{0,1\}^N} \left(\sum_{i \in \mathcal{E}} b_i x_i - \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} p_{ij} x_i x_j \right). \tag{1}$$

Since QUBO solvers minimize energy, we negate the objective to obtain the Hamiltonian component $H_{\text{obj}}(x)$:

$$H_{\text{obj}}(x) = -\sum_{i \in \mathcal{E}} b_i x_i + \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} p_{ij} x_i x_j. \tag{2}$$

Step 2 – Constraint. The requirement to select exactly K enzymes is expressed as the equality constraint:

$$\sum_{i \in \mathcal{E}} x_i = K. \tag{3}$$

Step 3 – Penalty Term. We enforce the constraint by adding a quadratic penalty term $H_{\text{pen}}(x)$ weighted by $\lambda > 0$:

$$H_{\text{pen}}(x) = \lambda \left(\sum_{i \in \mathcal{E}} x_i - K \right)^2. \tag{4}$$

Expanding the square and using the idempotent property $x_i^2 = x_i$ for binary variables:

$$\begin{aligned} H_{\text{pen}}(x) &= \lambda \left(\sum_{i \in \mathcal{E}} x_i^2 + \sum_{\substack{i,j \in \mathcal{E} \\ i \neq j}} x_i x_j - 2K \sum_{i \in \mathcal{E}} x_i + K^2 \right) && (5) \\ &= \lambda \left(\sum_{i \in \mathcal{E}} x_i + 2 \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} x_i x_j - 2K \sum_{i \in \mathcal{E}} x_i + K^2 \right) && (6) \\ &= \sum_{i \in \mathcal{E}} \lambda(1 - 2K) x_i + \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} 2\lambda x_i x_j + \lambda K^2. && (7) \end{aligned}$$

Step 4 – Combined Hamiltonian. The total QUBO Hamiltonian $H(x) = H_{\text{obj}}(x) + H_{\text{pen}}(x)$ is:

$$\begin{aligned} H(x) &= -\sum_{i \in \mathcal{E}} b_i x_i + \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} p_{ij} x_i x_j + \sum_{i \in \mathcal{E}} \lambda(1 - 2K) x_i + \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} 2\lambda x_i x_j + \lambda K^2 && (8) \\ &= \sum_{i \in \mathcal{E}} \left(\lambda(1 - 2K) - b_i \right) x_i + \sum_{\{i,j\} \subseteq \mathcal{E}, i<j} \left(p_{ij} + 2\lambda \right) x_i x_j + \lambda K^2. && (9) \end{aligned}$$

Step 5 – Q Matrix and Offset. The QUBO is defined by a symmetric matrix Q and an offset c such that $H(x) = x^\top Q x + c$. Reading off the coefficients from the general expression above:

$$\begin{aligned} Q_{i,i} &= \lambda(1 - 2K) - b_i, \quad \forall i \in \mathcal{E}, && (10) \\ Q_{i,j} &= p_{ij} + 2\lambda, \quad \forall \{i,j\} \subseteq \mathcal{E}, i < j, && (11) \\ Q_{j,i} &= p_{ij} + 2\lambda, \quad \forall \{i,j\} \subseteq \mathcal{E}, i < j, && (12) \\ c &= \lambda K^2. && (13) \end{aligned}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that infeasible solutions have strictly higher energy than any feasible solution. Let $B_{\max} = \max_{i \in \mathcal{E}} b_i$. The maximum possible gain in the original objective function from violating the cardinality constraint is bounded by $B_{\max}$ (e.g., by selecting an extra enzyme with the highest benefit). Therefore, choosing $\lambda > B_{\max}$ guarantees that the penalty cost for any deviation from $\sum x_i = K$ exceeds the maximum potential improvement in the objective. A safe concrete choice for testing is $\lambda = 100$ when all benefit scores are less than 100.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0,1\}^N$ recovers the optimal solution because $H(x)$ is constructed as the sum of the negated objective and a non-negative penalty term that vanishes if and only if the constraint is satisfied. For any feasible solution x satisfying $\sum_{i\in\mathcal{E}} x_i = K$, $H(x)$ equals the negated objective value. For any infeasible solution, $H(x)$ includes a strictly positive penalty term. Since $\lambda > B_{\max}$, the energy of any infeasible solution exceeds that of the optimal feasible solution. Consequently, the global minimum of $H(x)$ must occur at a feasible assignment that minimizes the negated objective, which corresponds to maximizing the original total score.

Illustrative Example. Consider a small instance with $N = 3$ enzymes, indexed $0, 1, 2$. Let the benefit scores be $b_0 = 10$, $b_1 = 20$, and $b_2 = 15$. Let the redundancy penalties be $p_{01} = 5$, $p_{02} = 2$, and $p_{12} = 8$. We require selecting exactly $K = 2$ targets. We choose $\lambda = 100$, which satisfies $\lambda > \max\{10, 20, 15\}$.

Substituting these values into the general formulas:

$$Q_{0,0} = 100(1-4) - 10 = -310, \tag{14}$$
$$Q_{1,1} = 100(1-4) - 20 = -320, \tag{15}$$
$$Q_{2,2} = 100(1-4) - 15 = -315, \tag{16}$$
$$Q_{0,1} = 5 + 200 = 205, \tag{17}$$
$$Q_{0,2} = 2 + 200 = 202, \tag{18}$$
$$Q_{1,2} = 8 + 200 = 208, \tag{19}$$
$$c = 100 \cdot 2^2 = 400. \tag{20}$$

The resulting QUBO Hamiltonian is:

$$H(x) = -310x_0 - 320x_1 - 315x_2 + 205x_0x_1 + 202x_0x_2 + 208x_1x_2 + 400. \tag{21}$$

Evaluating $H(x)$ for the three feasible solutions (x with exactly two ones):

- $x = (1, 1, 0)$: $H = -310 - 320 + 205 + 400 = -25$. Original score $= -(H - c) = 425$? No, score $= -(H - c)$ is incorrect; score $= -(H - c)$ only if $H = -\text{Score} + \text{Pen}$. Here $H = -\text{Score} + \text{Pen}$. Score $= -(H - \text{Pen}) = -(H - 0) = -H$ for feasible. Score $= 25$.
- $x = (1, 0, 1)$: $H = -310 - 315 + 202 + 400 = -23$. Score $= 23$.
- $x = (0, 1, 1)$: $H = -320 - 315 + 208 + 400 = -27$. Score $= 27$.

The minimum energy is $H = -27$ at $x = (0, 1, 1)$, corresponding to the maximum total score of 27. This solution selects enzymes 1 and 2, yielding a benefit of $20 + 15 = 35$ and a penalty of 8, for a net score of 27.